Alignment, position-based, and hybrid models of locust collective motion: a four-metric calibration against field trajectories

Asher Albrecht George Stephenson
Department of Computer Science, University of Colorado Boulder

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Abstract

We test which local interaction rule reproduces the collective motion of free-ranging locust hopper bands. We simulate the Vicsek alignment model [7], a position-based pull model after Sayin et al. [6], and a sequence of hybrids that combine forward-weighted alignment, anisotropic noise, heading inertia, and a soft preferred-spacing zone. We compare each model to the trajectory dataset of Weinburd et al. [8] across four metrics: polarization Φ, turning-angle standard deviation, nearest-neighbor distance, and a global anisotropy index of the body-centered neighbor density. The two baseline models fail in distinct, mechanistic ways. Hand-tuning a hybrid closed some gaps but introduced others. We replace hand-tuning with the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) [4], parallelized across 24 CPU cores, and search a six- and seven-dimensional parameter space against a weighted relative-error loss. The calibrated six-parameter hybrid (v4) matches all three scalar field metrics within 10% and reproduces the field's body-centered neighbor density both globally ($A_{\text{global}} \approx 0$) and inside a 1.5–3.0 cm near-zone band ($A_{\text{near}} = -0.005$ for both v4 and the field). A seven-parameter variant adding temporal bearing integration (v5) reaches a marginally lower scalar loss but lands in a different basin where the preferred-spacing ring carries a forward-bias artifact ($A_{\text{near}} = -0.06$) that the field does not have; the seventh parameter is unnecessary at the field's measurement scale and worsens the spatial fit. The fit depends on periodic boundaries; under reflective walls the calibration breaks, which we discuss as a limitation for any claim about open hopper-band geometry.

1 Introduction

Australian plague locusts (*Chortoicetes terminifera*) form hopper bands that march coherently across kilometers of terrain, causing agricultural damage at continental scale. Two competing theories of how they coordinate have appeared in the literature. The first, originating with Vicsek et al. [7] and applied to locusts by Buhl et al. [2], proposes that each individual aligns its heading with neighbors inside a fixed radius. Order arises from a density-driven phase transition. The second, advocated by Sayin et al. [6], proposes that individuals are attracted to the positions of forward neighbors. The interaction is geometric rather than directional, and order is a byproduct of position tracking rather than heading averaging.

These theories make different predictions about the spatial structure of the swarm and about the frame-to-frame heading dynamics of individuals. Until recently they could not be tested against individual-level field data. Weinburd et al. [8] released the first such dataset: roughly 3.3 million position records and 20,000 trajectories from four hopper bands, sampled at 25 Hz. We use this dataset as the ground truth.

We ask which local rule, or which combination of rules, reproduces the field measurements. Our contribution is twofold. First, we show that both pure alignment and pure pull fail in characteristic ways and that a hand-tuned hybrid trades one failure for another. Second, we replace hand-tuning with derivative-free optimization: CMA-ES searches the parameter space against a multi-metric loss and finds a hybrid that fits the field on every metric we tested. The methodological pivot, from exploring a few parameter combinations to optimizing many jointly, is the main lesson of the work.

2 Background and related work

The Vicsek model [7] is a minimal self-propelled particle system: at each time step, every agent updates its heading to the arithmetic mean of neighbor headings inside an interaction radius, plus uniform angular noise of width η. Buhl et al. [2] fit this model to laboratory locust experiments and reported a phase transition from disorder to coherent marching as density increased. Bazazi et al. [1] proposed an alternative mechanism in which collisions and cannibalism drive forward motion. Neither study had access to free-ranging field trajectories.

Weinburd et al. [8] computed body-centered neighbor density maps from their hopper-band data and observed an asymmetric distribution: more density behind the focal individual than ahead, with the dominant signal in the near zone. They argued for directional interactions but stopped short of a specific mechanistic model.

Sayin et al. [6] fit a position-based model to laboratory and modeling experiments. In their pull formulation, each agent steers toward a weighted centroid of forward neighbors, with weights proportional to $\cos^2(\beta/2)$ for relative bearing β. They argued that this mechanism can produce coherent motion without explicit alignment.

The Couzin model [3] combines repulsion, alignment, and attraction in nested zones. We borrow the soft preferred-spacing idea from this literature for our hybrid extensions.

CMA-ES [4] is a derivative-free evolutionary optimizer that adapts a multivariate Gaussian sampling distribution as it explores. It is well suited to noisy black-box objectives in low-dimensional continuous spaces. We use the `pycma` implementation [5].

3 Methods

3.1 Field metrics

We compute four metrics from the Weinburd dataset and apply the same functions to every simulated trajectory:

1. Global polarization $\Phi(t) = \left|\frac{1}{N(t)} \sum_i e^{i\theta_i(t)}\right|$, averaged over time after burn-in.
2. Turning-angle standard deviation: per-step heading change $\Delta\theta_i(t) = \theta_i(t+1) - \theta_i(t)$, wrapped to $[-\pi, \pi]$, pooled across agents and time.
3. Median nearest-neighbor distance (NND).
4. Body-centered neighbor density. For each focal individual we rotate the relative position of every neighbor inside a 10 cm radius into the focal's body frame (heading along $+y$) and accumulate a 50×50 histogram. Figure 1 shows the field distribution.

From the density map we derive a scalar global anisotropy index $A = (\rho_{\text{rear}} - \rho_{\text{front}})/(\rho_{\text{rear}} + \rho_{\text{front}})$, where ρ_{rear} and ρ_{front} are the mean densities in the lower and upper halves of the body-centered histogram. $A > 0$ indicates more density behind the focal than ahead.

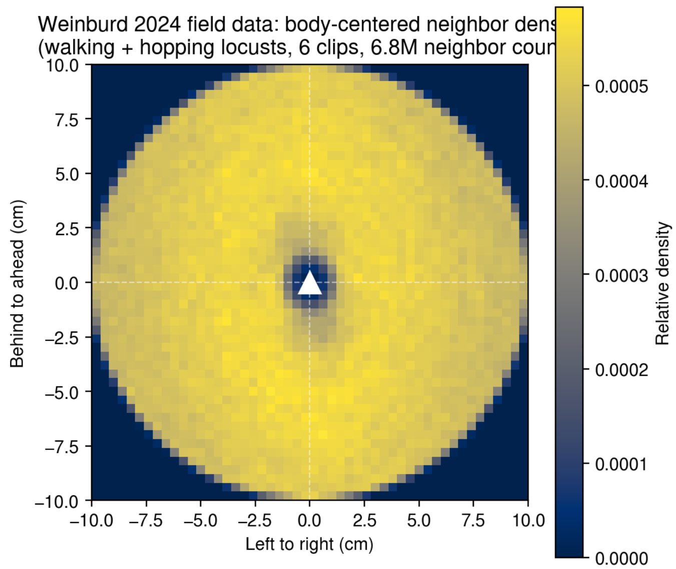

Figure 1: Body-centered neighbor density from the Weinburd 2024 hopper-band dataset, walking and hopping individuals combined (six clips, 6.86×10^6 neighbor counts). The focal heads up. The distribution is close to isotropic at the global level ($A_{\text{field}} = -0.0008$), with a self-exclusion hole at the origin. We use this map and its scalar summary as a fourth field-fitting target alongside the three scalar metrics.

3.2 Models

The Vicsek model implements standard alignment dynamics with periodic boundaries. Each agent sets its next heading to $\theta_i(t+1) = \arg\sum_{j\in\mathcal{N}_i} e^{i\theta_j} + \eta\xi_i$, where $\mathcal{N}_i$ is the set of neighbors within $r_{\text{int}} = 7$ cm and $\xi_i \sim \mathcal{U}(-1/2, 1/2)$. Speed is constant and a hard repulsion override applies at $r_{\text{rep}} = 1.5$ cm.

The pull model has each agent steer toward the centroid of forward neighbors weighted by $w(\beta) = \cos^2(\beta/2)$, following Sayin et al. [6]. The noise term and the repulsion override match the Vicsek implementation.

Hybrid v3 (anisotropic Vicsek) extends the alignment rule along two axes. Forward neighbors (relative bearing $|\beta| < \pi/3$) get weight $1 + \alpha_{\text{aniso}}$ and rear neighbors $1 - \alpha_{\text{aniso}}$. Effective noise scales with the fraction of neighbors ahead: $\eta_{\text{eff}}(i) = \eta_{\text{base}}(1 - \lambda f_{\text{forward}}(i))$. Speed varies with local order between $v_{\min}$ and $v_{\max}$.

Hybrid v4 adds two mechanisms motivated by the v3 failure analysis (Section 4). The heading-inertia term has each agent commit a fraction $\mu \in (0, 1]$ of the way to its new desired heading θ^* via exponential smoothing in unit-vector space: $\theta_i(t+1) = \arg\big[(1-\mu)e^{i\theta_i(t)} + \mu e^{i\theta^*}\big]$. This narrows the per-step turning distribution. The preferred-spacing zone applies in the band $r_{\text{rep}} < d < r_{\text{pref}}$: the agent mixes a radial-push direction into its desired heading with weight k_{pref}.

Hybrid v5 adds temporal bearing integration after Sayin et al. [6]'s ring-attractor analogy. Each agent maintains an exponential moving average of the alignment-derived target unit vector, $T_{\text{ema}} \leftarrow (1 - \rho)T_{\text{ema}} + \rho T_{\text{instantaneous}}$. The desired heading derives from T_{ema} rather than the instantaneous target. Setting $\rho = 1$ disables the integrator and recovers v4.

3.3 Calibration

We calibrate the hybrid models with CMA-ES against a weighted relative-error loss

$$L = w_\Phi \frac{|\Phi_{\text{sim}} - \Phi_{\text{field}}|}{\Phi_{\text{field}}} + w_\sigma \frac{|\sigma_{\text{sim}} - \sigma_{\text{field}}|}{\sigma_{\text{field}}} + w_{\text{NND}} \frac{|\text{NND}_{\text{sim}} - \text{NND}_{\text{field}}|}{\text{NND}_{\text{field}}}, \tag{1}$$

with weights $w_\Phi = 1$, $w_\sigma = w_{\text{NND}} = 2$. The two failure-mode metrics from the baseline experiments (Section 4) get double weight; polarization is already close in v3. Each evaluation runs five seeds of a 2000-step simulation with 500-step burn-in and returns the ensemble mean.

The optimization parameter space is six-dimensional for v4 and seven-dimensional for v5: $\eta_{\text{base}} \in [0.1, 1.5]$, $\lambda \in [0, 1]$, $\alpha_{\text{aniso}} \in [0, 1]$, $\mu \in [0.1, 1.0]$, $k_{\text{pref}} \in [0, 0.7]$, $r_{\text{pref}} \in [1.6, 5.0]$ cm, and (v5 only) $\rho_{\text{temporal}} \in [0.05, 1.0]$. Following the `pycma` guidance [5], we map every parameter linearly to the internal range $[0, 10]$ so a single σ_0 has uniform meaning across dimensions.

We run CMA-ES with population size 20, $\sigma_0 = 2.0$, and 40–50 generations. The 20 candidate evaluations within each generation are distributed across CPU cores using `multiprocessing.Pool`; BLAS threading inside workers is pinned to one to avoid oversubscription. On a 32-thread Intel i9-13900K the full v4 optimization completes in about 13 minutes (gen_0 in 19 seconds, end-to-end speedup of about $11\times$ over the serial implementation).

4 Results

4.1 The two baselines fail in distinct ways

After calibrating η to match field polarization, the Vicsek model hits $\Phi = 0.82$ (target 0.820) but produces a turning-angle distribution more than twice as wide as the field (0.595 rad vs 0.276 rad, 116% relative error) and undercounts NND by 33%. The model achieves order, but at the cost of pathological heading noise that compensates for the absence of any heading-stabilizing mechanism beyond raw alignment.

The pull model fails differently. Across five structural variants we tested (varying cone half-angle, pull strength, and noise), polarization remained at $\Phi \approx 0.06$, with a turning-angle distribution near 2 rad (essentially saturated). The geometric reason is that, in a disordered swarm, neighbor bearings are uniformly distributed around the focal, and the forward-weighting function $\cos^2(\beta/2)$ is symmetric under reflection. The expected pull direction is along the focal heading, with no symmetry breaking that would amplify small alignment fluctuations into global order.

Hybrid v3 (anisotropic Vicsek with hand-tuned $\eta = 1.1$, $\lambda = 0.8$, $\alpha = 0.6$) hits polarization within 0.5% but still misses turning-angle std by 215% and NND by 55%. Forward weighting tightens the alignment, which lowers η enough to bring polarization in at a sensible noise level, but introduces no mechanism that limits per-step heading change or sets a preferred spacing.

4.2 CMA-ES closes the scalar gaps

Figure 2 shows the loss trajectories. The v4 run reaches a final loss of 0.099, with $\Phi = 0.898$ (9.6% off the field), turning-angle std $= 0.276$ rad (0.001% off), and NND median $= 3.899$ cm (0.2% off). The two failure modes that motivated heading inertia and preferred spacing close to within seed noise. Polarization remains slightly elevated.

The v5 run reaches 0.085, but the converged $\rho_{\text{temporal}} = 0.95$ is at the no-effect end of its allowed range. The improvement comes from the optimizer escaping a 6-D local minimum into a different basin, not from using the new mechanism. Specifically, α_{aniso} saturates at 1.0 and λ drops to 0.17, while

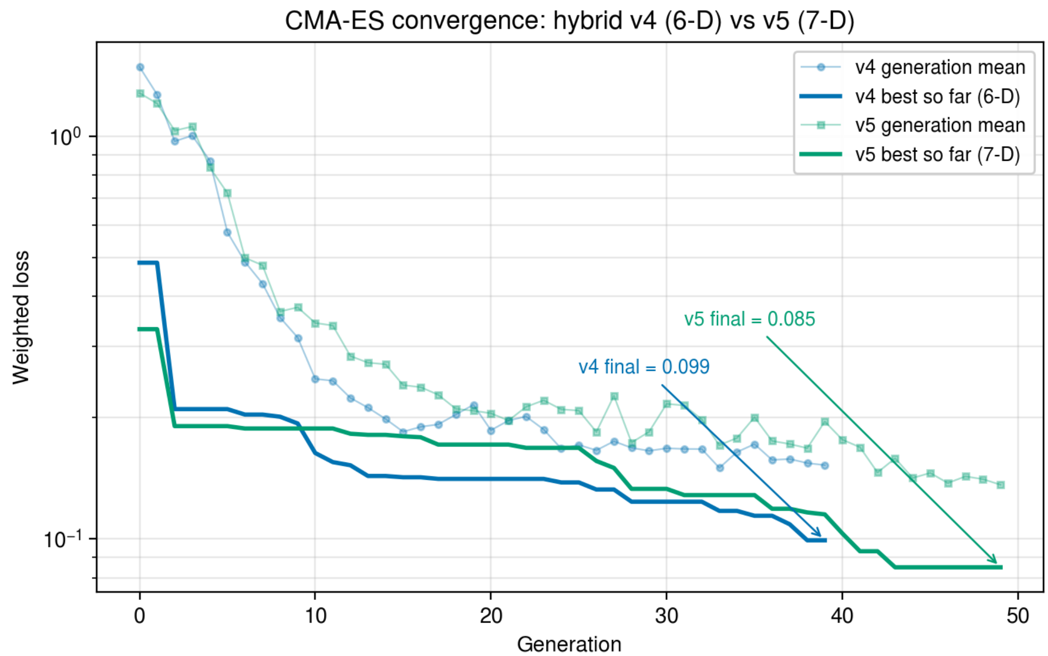

Figure 2: CMA-ES convergence curves. Both v4 (six-dimensional) and v5 (seven-dimensional) reduce the weighted loss by an order of magnitude over 40–50 generations. The v5 run plateaus around generation 25 before a late breakout finds a slightly better basin.

the other parameters move only slightly. The basin v5 escaped into is worse on the spatial metric we introduce in Section 4.3, so the marginally lower scalar loss does not generalize.

Figure 3 summarizes per-metric performance for every model we ran. Numerical values are listed in Table 1. The two CMA-ES-calibrated runs separate further once the spatial structure is examined directly (Section 4.3): v4 reproduces the field's near-zone density, v5 does not.

4.3 Spatial structure: v4 matches, v5 distorts

The body-centered neighbor density map is the test that motivated the hybrid in the first place. Weinburd et al. [8] reported anisotropic neighbor distributions in their data, and our hybrid v3 reproduced a visible asymmetry in the inner ring of its density map.

The global anisotropy index is essentially zero in the field ($A_{\text{global}} = -0.0008$). v3, v4, and v5 all reproduce the global value to within $|A| < 0.001$ (Figure 4, panel titles). The original "forward void" claim does not hold at the field scale.

Restricting A to a 1.5–3.0 cm radial band separates the calibrated hybrids. The near-zone index A_{near} is -0.005 in the field, -0.003 in v3, and -0.005 in v4 (all close to flat). v5's value is -0.061, an order of magnitude larger, with more density ahead of the focal than behind. The preferred-spacing ring at v5's optimum sits closer to the focal in front than at the rear, at radii where the field shows no asymmetry. The white dotted circles in Figure 4 mark the band.

The CMA-ES loss did not include A_{near}, so the optimizer could place near-zone structure anywhere that did not break the three scalar metrics. v4 found a basin in which the preferred-spacing ring sits symmetrically around the focal ($\alpha_{\text{aniso}} = 0.80$, $\lambda = 0.52$). v5, with the extra ρ_{temporal} dimension, found a different basin: α_{aniso} saturates at 0.99 and λ drops to 0.17, which sharpens the ring and biases it forward. v5's scalar loss is 0.085 versus v4's 0.099, so the optimizer prefers it; on the spatial metric the

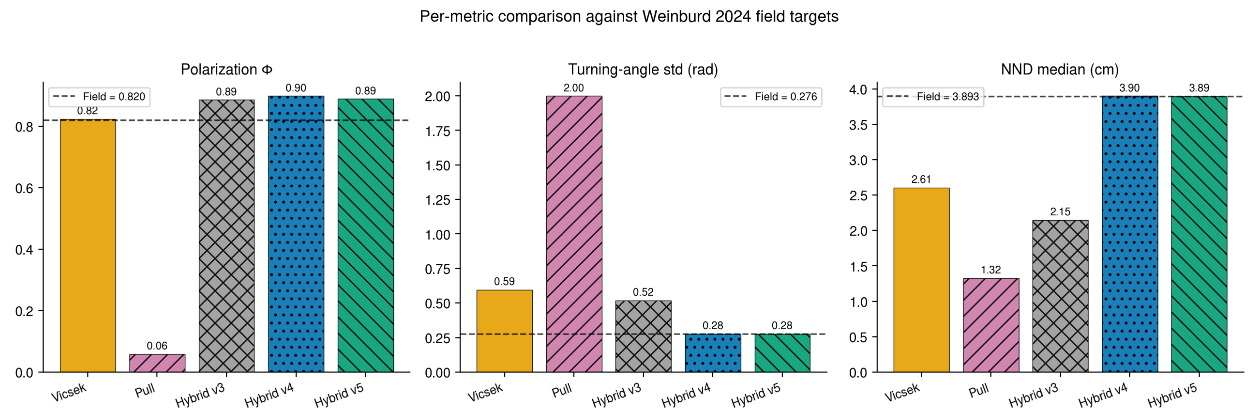

Figure 3: Field-fit scorecard across all models on the three scalar metrics. Dashed lines indicate field targets. Vicsek and Pull miss multiple metrics. Hybrid v3 closes polarization but not the others. CMA-ES-calibrated v4 and v5 hit all three scalar targets within seed noise. The body-centered spatial structure separates v4 from v5; see Figure 4.

Table 1: Field-fit scorecard. Relative error in parentheses. Field targets: $\Phi = 0.820$, $\sigma = 0.276$ rad, NND $= 3.893$ cm. $\pm$ values on Φ are the across-seed standard deviation of the per-seed mean (10 seeds for the baseline runs, 5 seeds for the CMA-ES-calibrated runs).

Model	$\Phi \pm \sigma_{\text{seed}}$	σ (rad)	NND (cm)
Vicsek	0.824 ± 0.019 (0.5%)	0.595 (116%)	2.61 (33%)
Pull	0.058 ± 0.001 (93%)	1.999 (624%)	1.32 (66%)
Hybrid v3	0.890 ± 0.025 (8.5%)	0.520 (88%)	2.15 (45%)
Hybrid v4	0.898 ± 0.027 (9.6%)	0.276 (0.001%)	3.90 (0.2%)
Hybrid v5	0.889 ± 0.012 (8.4%)	0.276 (0.05%)	3.89 (0.04%)

trade goes the other way. The seventh parameter is unnecessary at the field's measurement scale, as Section 4.2 already noted, and now actively worsens the spatial fit. v4 is the configuration that matches all four metrics; v5 is a worse fit dressed up as a slightly better one.

The "void emerges" claim from earlier v3 figures was an overinterpretation of a feature that is small in the field. v3 reproduces the field's near-zone structure for the same reason it fails the scalar metrics: forward-weighted alignment alone is enough for spatial structure but not for tight heading control or NND.

4.4 Boundary-condition dependence

We tested the calibrated v4 model under reflective walls instead of periodic boundaries. The fit broke: turning-angle std jumped to 0.493 rad (78% off) and NND dropped to 3.47 cm (11% off). Polarization actually moved closer to target, from 0.898 to 0.752 (8.3% off). The periodic image structure is doing real physical work in the calibration. Reflective walls perturb headings on contact, inflating the per-step turning distribution, and they compress densities away from the boundary, lowering NND.

This matters because hopper bands are open systems with front and rear asymmetry, not periodic boxes. A model intended to claim something about field hopper-band dynamics needs to be calibrated under realistic boundaries from the start, not validated against them after the fact. We flag this as a limitation rather than a fix.

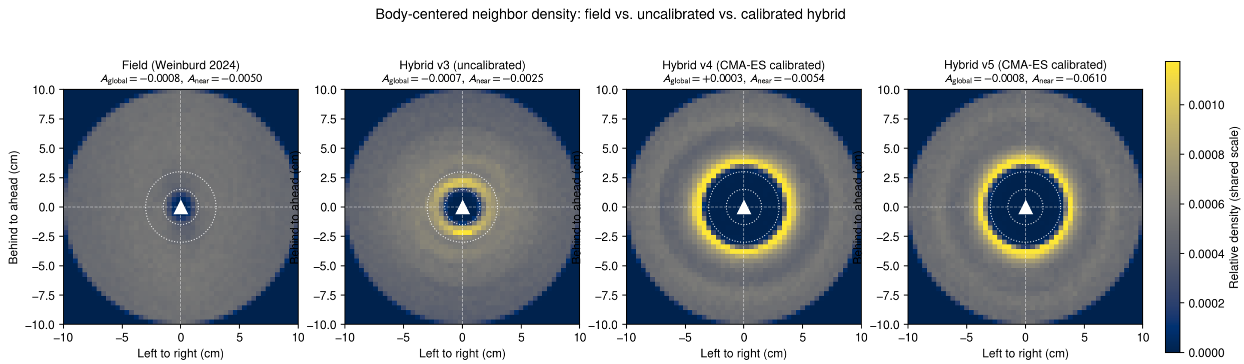

Figure 4: Body-centered neighbor density for the field and the three hybrid models, on a shared color scale. The global anisotropy index is within $|A| < 0.001$ of zero in all four. The near-zone index, computed only inside the white dotted band ([1.5, 3.0] cm), separates the v5 ring artifact ($A_{\text{near}} = -0.061$) from the near-flat field (-0.005), v3 (-0.003), and v4 (-0.005). v4 fits both scalar and near-zone metrics; v5 trades the spatial fit for a marginally lower scalar loss.

5 Discussion

We spent several weeks hand-tuning a hybrid model and producing scorecards that captured one metric well at a time. CMA-ES, with parallel evaluation across cores, runs end-to-end in about 13 minutes and finds parameter combinations that hand-tuning missed. For noisy black-box objectives in this dimensionality, derivative-free evolutionary search is the right default. We should have used it sooner.

Adding temporal bearing integration (ρ_{temporal}) gave the optimizer a seventh degree of freedom that it drove to its no-effect limit ($\rho = 0.95$). On the scalar loss alone the change looks benign and even slightly helpful, but the extra dimension also shifted the converged basin: α_{aniso} saturated at 1.0 in v5, against v4's 0.80, and that re-shaped the preferred-spacing ring into the forward-biased configuration that breaks the near-zone fit (Section 4.3). The mechanism is not wrong in principle, but at the field's measurement scale it is unnecessary and, in our calibration, harmful. Future work that tests Sayin et al. [6]'s ring-attractor proposal will need a measurement that distinguishes instantaneous from time-integrated bearing estimation, which neither our scalar nor our spatial metrics do.

The remaining gap is polarization. The v4 and v5 calibrations sit at $\Phi \approx 0.89$ versus a field target of 0.82. We re-ran v4 with equal weights ($w_\Phi = w_\sigma = w_{\text{NND}} = 1$) to test whether the gap was an artifact of our prioritized weighting. The equal-weight optimum lands at $\Phi = 0.883$, $\sigma = 0.277$ rad, NND $= 3.92$ cm, with λ dropping from 0.52 to 0.24 and α_{aniso} saturating at 0.99. The polarization gap narrows by 0.015 and the other two metrics do not break. v4's polarization floor is therefore close to 0.88 regardless of weighting, which we read as a structural property of the model at this density rather than a tunable parameter.

We have not run an open-boundary calibration. The reflective-wall stress test indicates the trade-offs differ. Three plausible follow-ups are: a flux-boundary scheme that despawns leaving agents and respawns at the rear; calibration under reflective walls with the loss reweighted; or a sliding-window simulation centered on the swarm centroid. We did not have time to implement any of these.

6 Project contributions

Asher Albrecht built the field-data ingestion pipeline from the Weinburd et al. [8] `.mat` files, computed the four baseline metrics on the field data (Section 3.1), implemented the Vicsek and Pull baseline models, and led the hybrid v3 anisotropic-Vicsek design and the η/λ parameter sweep figures.

George Stephenson extended the hybrid model with heading inertia, preferred spacing, and temporal bearing integration (v4 and v5 in `hybrid_v4.py` and `hybrid_v5.py`), built the parallel CMA-ES calibration framework (`calibrate_cma.py`, `calibrate_cma_v5.py`, `validate_calibrated.py`), implemented the spatial-anisotropy validation against the field density map, and produced the post-calibration figures.

Both authors contributed to the experimental design, the slide deck, and this paper.

Code and data

The simulation code, calibration scripts, CMA-ES log files, and figure generators are at https://github.com/*(repo to be added on submission)*. The Weinburd dataset is available from the original authors at Weinburd et al. [8].

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